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Unit – 5 : Session – 1 : SLO – 2 | Modelling Simple Regular Expressions Using Automata

■ I. Questions & Answers

1. Regular expression for strings ending with 'ab' — $(a + b)^*ab$ i.e. any string over $\{a, b\}$ followed by the suffix ab.
2. DFA for a^* — a single state q_0 that is both the start and the accepting state, with a self-loop on 'a'. The empty string is accepted immediately, and every additional 'a' loops back to q_0 .
3. Converting a regular expression to an NFA — use Thompson's construction: build small NFAs for the basic symbols, then compose them with ϵ -transitions for union ($|$), concatenation, and Kleene star ($*$), yielding an ϵ -NFA equivalent to the expression.